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Luca Podavini

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ABOUT MYSELF

Artificial Intelligence Systems student at the University of Trento, specialized in Intelligent Robots. My expertise spans from Reinforcement Learning and Motion Planning to Real-time Computer Vision and Embedded Systems. I bridge the gap between high-level AI algorithms and low-level control, with a strong focus on C++, Rust, and ROS/ROS2.

Please visit my website (lucapodavini.it) for a more interactive and easy to read bio. You will also find the complete project portfolio with references to the technical reports, GitHub repositories and demos.

SKILLS

C/C++ | Rust | Python | PyTorch | ROS/ROS 2 | Robot Planning | Reactive Control | Optimal Control | Deep Learning | Reinforcement Learning | Computer Vision | Embedded Software | Docker | Git

LANGUAGE SKILLS

Mother tongue(s): Italian

English

LISTENING: B2 **READING:** B2 **WRITING:** B2

SPOKEN PRODUCTION: B2

SPOKEN INTERACTION: B2

WORK EXPERIENCE

Think Easy s.r.o - Czechia, Prague

Programmer

[09/2019 - 09/2019]

Gained hands-on experience in back-end web development using Django (Python) and front-end development using React (JS) during a one-month internship abroad.

Cualeva s.r.l - Italy, Puegnago Del Garda

Programmer

[12/2019 - 09/2022]

Started as a school internship and quickly evolved into a part-time position alongside my final high school year. Transitioned to a full-time role during the 2021 and 2022 summer breaks, eventually stepping down to focus entirely on my academic path at the university.

Developed internal functionalities for company products using .NET Framework (C#). Contributed to the development and maintenance of web platforms, desktop applications, and mobile apps.

EDUCATION & TRAINING

Master degree in Artificial Intelligence Systems

University of Trento [09/2024 - Current]

City: Trento | **Country:** Italy

Website: www.unitn.it | **Field(s) of study:** Intelligent Robots: | **Level in EQF:** 7

In-depth exploration of core AI branches, including Deep Learning, Computer Vision, and Natural Language Understanding, followed by a specialization in the Intelligent Robots path. Acquisition of advanced skills in robot control, ranging from foundational techniques (Reactive and Optimal Control) to state-of-the-art methodologies (Reinforcement Learning, Motion Planning, and Task Planning). Direct application of theoretical concepts on physical and simulated robots, as detailed in the project portfolio.

Bachelor's degree in Computer Science

University of Trento [09/2021 - 17/09/2024]

City: Trento | **Country:** Italy

Website: www.unitn.it/ | **Field(s) of study:** Information and Communication Technologies: | **Final grade:** 98/110 | **Level in EQF:** 6 | **Thesis:** Move Fuzzer

Development of a strong computer science foundation, with a focus on embedded software development and low-level programming (C, C++, Rust). The rigorous application of advanced data structures established a solid groundwork for subsequent robotics studies. The degree culminated in the exploration of low-level programming language mechanics through the thesis project "Move Fuzzer", an automated software testing tool.

Diploma in Information and Communication Technologies

I.I.S Luigi Cerebotani [09/2016 - 26/06/2021]

City: Lonato Del Garda | **Country:** Italy

Website: https://iislonato.edu.it/ | **Field(s) of study:** Information and Communication Technologies: | **Final grade:** 100/100 | **Level in EQF:** 4

Acquisition of practical knowledge in network architectures and IT infrastructure. Initial hands-on experience in web development, with a strong technical emphasis on back-end logic and relational database design (SQL).

PROJECTS

[10/2025 - 01/2026]

Optimization & Learning for Robot Control Project divided into three assignments to cover the following topics:

1. **Reinforcement Learning**: PPO algorithm applied to a unicycle navigation task in an obstacle-filled environment.
2. **Optimal Control**: applied both trajectory tracking and path tracking optimization techniques to a UR5 industrial robotic arm.
3. **Reactive Control**: reactive control framework to enable a humanoid robot to track complex trajectories and perform walking motions while maintaining dynamic balance.

Reinforcement Learning, PPO Algorithm, REINFORCE algorithm, Unicycle Kinematics, Trajectory Tracking, Navigation, Obstacle Avoidance, LIDAR, Optimal Control, UR5 Manipulator, Trajectory Optimization, Reactive Control, Task Space Inverse Dynamics (TSID), Humanoid Robotics

Links: www.lucapodavini.it/#unicycle-rl | www.lucapodavini.it/#ur5-optimal-control | www.lucapodavini.it/#humanoid-tsid

[01/2026 - 02/2026]

Robot Planning Motion planning project on an Agile X LIMO. Explored both Combinatorial and Sampling Based planning to solve the Victim Rescue problem in a controlled environment.

PRM, Approximate Cell Decomposition, Multipoint Dubins, Orienteering, Polygon Inflation, Shortcutting, ROS, RViz, Gazebo, Loco Nav, Simulation, Agile X LIMO, Victim Rescue

Link: www.lucapodavini.it/#robot-planning

[01/2026 - 02/2026]

Automated Planning Implemented different planning techniques such as classical planning, temporal planning, or hierarchical task networks. Compared the performance of different solvers to discover their peculiarities.

Classical Planning, Temporal Planning, Hierarchical Task Networks, PDDL, HDDL

Link: www.lucapodavini.it/#automated-planning

[05/2025 - 07/2025]

Deep Learning: Parameter Efficient Fine Tuning for CLIP This project explores Parameter Efficient Fine Tuning (PEFT) techniques applied to the CLIP model for image classification tasks. We implemented and compared several PEFT methods, including CoCoOp, CLIP-LoRA, and DISEFT, to evaluate their performance in few-shot learning scenarios.

Zero-Shot CLIP, CoCoOp, CLIP-LoRA, DISEFT, Few-Shot Learning, PyTorch, Generative AI

Link: www.lucapodavini.it/#deep-learning

[04/2025 - 06/2025]

Natural Language Understanding This project focuses on Language Modelling and Natural Language Understanding (NLU). Various architectures have been compared, including LSTM-based models with dropout layers and advanced optimization techniques like AdamW and NT-AvSGD, as well as transformer-based models such as BERT.

LSTM, Dropout Layers, AdamW, NT-AvSGD, Weight Tying, Variational Dropouts, Transformer, BERT, Intent Classification, Slot Filling, PyTorch

Link: www.lucapodavini.it/#nlu

[10/2023 - 04/2024]

CLAW MACHINE Designed and built an IoT claw machine from scratch, including custom aluminum framing, 3D printed parts, electronics, and full Embedded C software architecture.

This project was presented at Fablab Trento during ICT Days 2024.

Embedded C, MSP-432 (ARM Cortex M4), Bluetooth Control, LCD Display, Stepper Motors, Servo, Finite State Machine, 3D Printing, 3D Modelling

Link: www.lucapodavini.it/#claw-machine

[08/2025 - Current]

VISUAL-HIVE VJ software developed as a personal project to meet the specific needs of live events, prioritizing simplicity and real-time performance. It enables seamless overlaying of foreground logos onto dynamic background loops, with integrated effects like strobe and beat-synced motion.

C++, Open CV, MIDI Protocol, Ableton Link, Multithreading, Real-time Data Processing, Low-latency Visualization

Link: www.lucapodavini.it/#visual-hive